ZYNTH

Complete Application Blueprint v1.0

Synthesize yourself. Reach your zenith.

> *"You don't consume Zynth. You perform in it. The system watches, adapts, and evolves with you."*

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1. SYSTEM OVERVIEW & PHILOSOPHY

Zynth is not a learning platform. It is a **behavioral synthesis engine** — a system that observes how you show up every day, builds a living model of who you are, adapts everything to your patterns, and makes your own evolution visible in real time.

Core Philosophy

```
Static platforms deliver content.
Zynth delivers evolution.

The user does not consume. They perform.
The system does not teach. It synthesizes and reflects.
```

What Zynth IS and IS NOT

Zynth IS	Zynth IS NOT
A behavioral synthesis engine	A course or content platform
An adaptive identity system	A quiz or assessment tool
A gamified growth ecosystem	A habit tracker
A living character progression system	A certification platform
A competition and ranking platform	A social network

The Master Loop

The Golden Rule

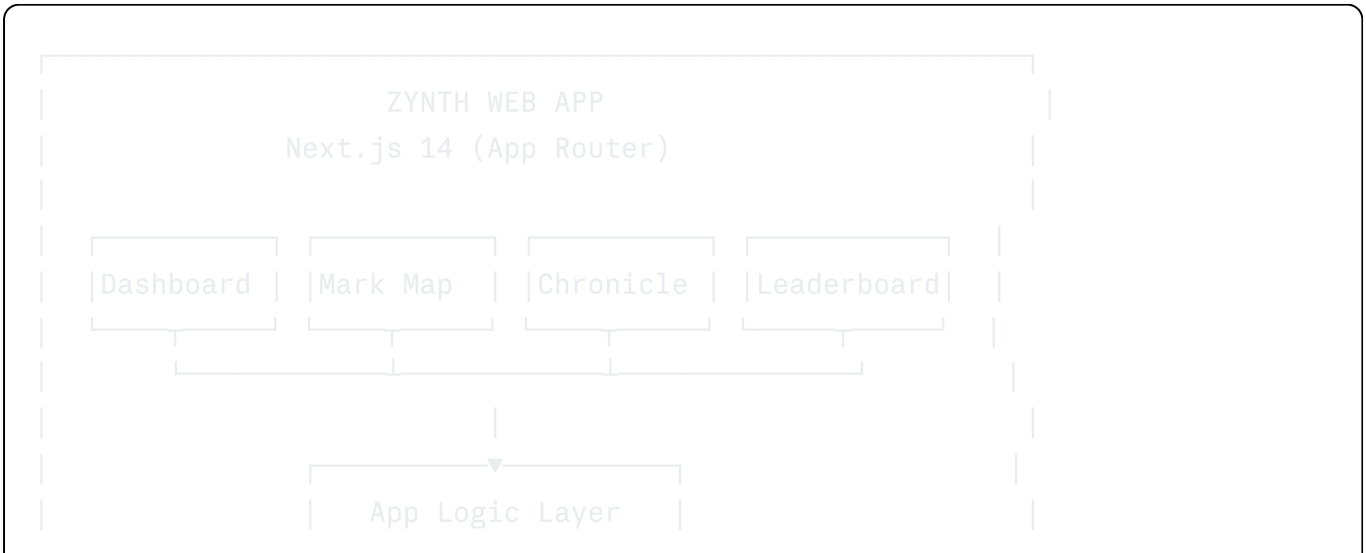
> **The LLM generates content and signals. The app owns all math. Numbers never**

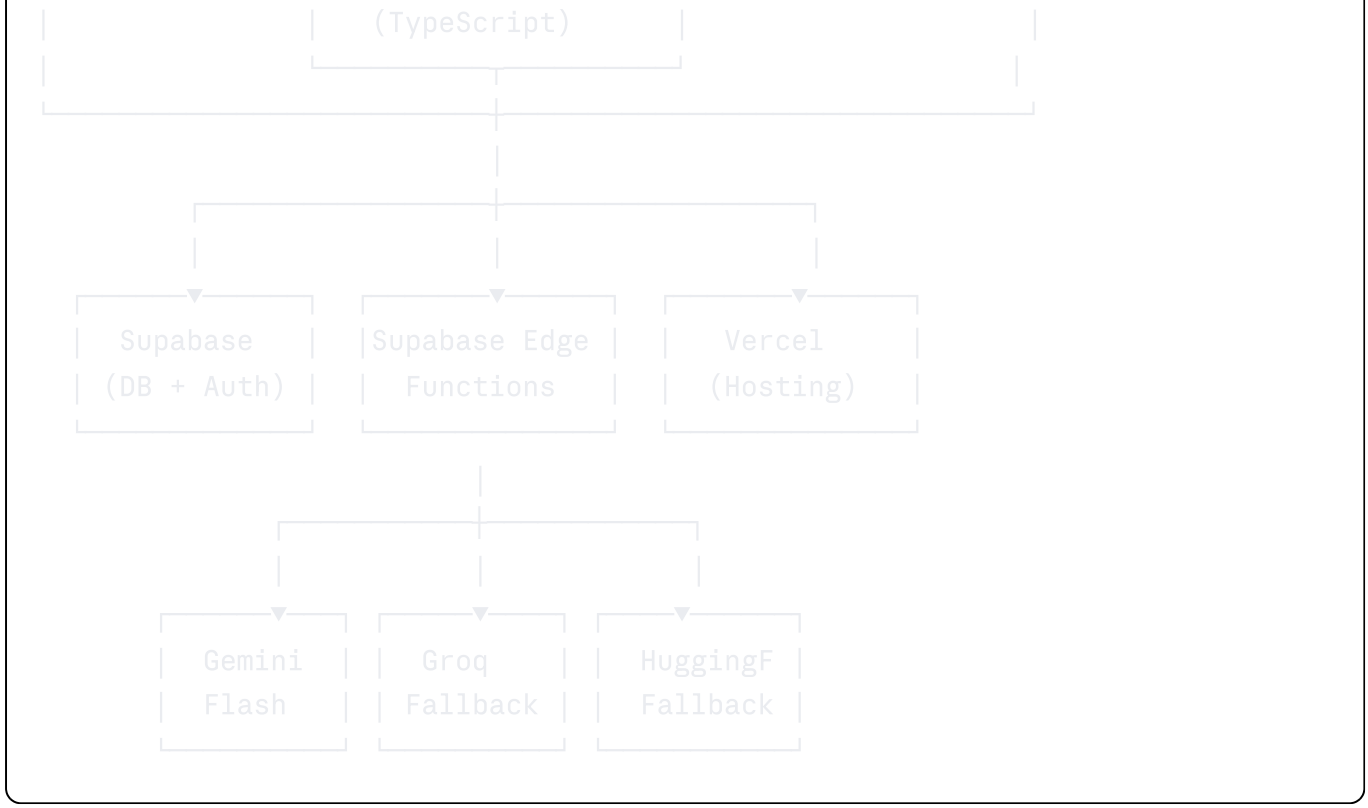
> **touch the LLM.**

Responsibility	Owner
Quest content generation	LLM
Response grading (quality judgment)	LLM
Behavioral signal extraction	LLM (strings only)
Skill evolution naming	LLM
Hidden skill discovery messages	LLM
Daily profiling question generation	LLM
EXP calculation	App
Stat adjustments	App
Level-up check	App
Streak tracking	App
Token & XP card management	App
Achievement checking	App
Anti-exploit detection	App
Leaderboard calculation	App
All number changes	App — NEVER LLM

2. APP ARCHITECTURE

Stack





LLM Fallback Chain

```
Request
   |
   v
Gemini Flash ──► success ──► respond
   |
   fail / rate limit
   |
   v
Groq (Llama 3.3 70B) ──► success ──► respond
   |
   fail
   |
   v
Hugging Face API ──► success ──► respond (simple tasks only)
   |
   fail
   |
   v
Local fallback quest cache ──► serve cached quest
```

---

3. SCREEN MAP & USER FLOWS

Navigation Structure

```
MAIN NAV (5 tabs always visible except in immersive mode):
[ Dashboard ] [ Mark Map ] [ Chronicle ] [ Leaderboard ] [ Profile ]
```

```
IMMERSIVE SCREENS (nav disappears):
Quest Screen → Grade Reveal → Event Screens (queue)

OVERLAY SCREENS:
Daily Pop-up (energy + profiling question)
Settings (from Profile)
```

Full Screen Map


```
SPECIAL:
[PROFESSION EVOLUTION] — appears on login when triggered
[PRESTIGE SCREEN] — from Profile, Level 100+
[ACHIEVEMENT PAGE] — from Profile
[SETTINGS] — from Profile
```

Slide Transition System

```
Dashboard → Mark Map:
  Dashboard content slides OUT to the LEFT
  Mark Map faintly visible in background BECOMES DOMINANT
  Mark Map content slides IN from the RIGHT
  Animation: 350ms ease-in-out, simultaneous

Mark Map → Dashboard:
  Reverse of above — Mark Map slides right, Dashboard returns from left

All other tab transitions: standard cross-fade (200ms)
Quest Screen: slides up from bottom, covers everything including nav
Grade Reveal: fades in over Quest Screen
Event Screens: full page replacements with themed animations
```

---

4. ENGINE 1: ONBOARDING ENGINE

Purpose

Convert 5 oblique psychological scenario questions into a fully structured player profile. Questions feel like a personality quiz — never reference learning, studying, or skills. The user reveals more because they don't know what they're revealing.

Flow

```
User Answers Q1–Q5
        |
        ▼
  Answer Collector
  (App formats JSON)
        |
        ▼
  LLM Classification
  (Gemini Flash via Edge Function)
        |
        ▼
  Parse + Validate
  (Zod schema check)
        |
        ▼
  App Appends Scaffolding
  (level, exp, tokens, etc.)
        |
        ▼
  Write to Supabase → profiles table
        |
```

Route to Profile Reveal screen

The 5 Questions

Q1 — Maps: Domain + Execution pattern

"You have $10,000 land in your account with a note: 'spend it in 48 hours or lose it.' What actually happens?"

Options:

- A — I buy tools, equipment, or resources for something I've been wanting to build
- B — I research obsessively for 47 hours and spend it in the last hour on exactly the right thing
- C — I invest it into something that will grow — money should make more money
- D — I create an experience — travel, document everything, make it unforgettable
- E — I split it between people I care about and something that matters to me
- F — I freeze. 48 hours passes. I probably lose it.

Signal map: A=Technical/builder+high execution, B=Analytical+overthinking weakness, C=Strategic+long-term, D=Creative+experiential, E=Social domain, F=Low execution+fear_of_failure

---

Q2 — Maps: Learning style + Behavior signature

"A fire alarm goes off in a building you're in. False alarm. What were you doing before it pulled you away?"

Options:

- A — Making or building something — I was deep in it and annoyed at the interruption
- B — Reading, watching, or consuming something that caught my attention earlier
- C — Talking to someone — it was actually a good conversation
- D — Organizing something, planning, mapping out an idea in my head or on paper
- E — On my phone, not doing much — I needed a break anyway
- F — Working toward something I set for myself — I was in a rare focused state

Signal map: A=Practical learner+Technical/Creative, B=Analytical learner+curiosity, C=Social+Guided learner, D=Guided+Strategic, E=Inconsistency weakness, F=Exploratory+high consistency

---

Q3 — Maps: Hidden strengths + Weakness signal + Domain

"If you could wake up tomorrow with one of these and use it every single day — which one do you actually choose?"

Options:

- A — The ability to learn any skill perfectly in one focused hour
- B — Knowing exactly why anything works the way it does, instantly
- C — Whatever I imagine comes out perfectly in real life — exactly as I pictured it
- D — Anyone I speak to understands me completely, no miscommunication ever
- E — I can see the single best decision to make in any situation before I make it
- F — I never forget anything I've ever experienced or learned

Signal map: A=Execution-driven+Practical, B=Analytical+insight, C=Creative+imagination, D=Social+communication, E=Strategic+decision, F=Analytical+memory_crutch weakness

---

Q4 — Maps: Consistency + Execution + Weakness signal

"Completely free week. No obligations, no people expecting anything. By day 3 — what's honestly happening?"

Options:

- A — I'm deep in something I started on day 1 and I haven't come up for air
- B — I've gone down five different rabbit holes and I'm somehow more tired than before
- C — I made a loose plan on day 1 and I'm roughly sticking to it, feels good
- D — Day 1 I had big plans. Day 3 I'm horizontal watching something I've already seen
- E — I've somehow ended up organizing things around other people — plans, hangouts, something social
- F — I started four things. Finished none. Classic.

Signal map: A=High consistency+execution+flow, B=Exploratory+scattered_interests weakness, C=Guided+moderate consistency, D=Poor_focus+low execution, E=Social+externally motivated, F=Inconsistency weakness+starter-not-finisher

---

Q5 — Maps: Behavior signature + Self-awareness + Social signals

"Someone who genuinely knows you is describing you to a stranger. Which one lands closest to what they'd actually say?"

Options:

- A — "Give them a problem. They'll figure it out — they always do somehow"
- B — "They know a little about everything and a lot about whatever they're obsessed with this month"
- C — "Full of ideas. Whether they do anything with them depends on the day"
- D — "Quietly gets things done. You won't hear about it until it's finished"

- E — "Brilliant but their own biggest obstacle — they overthink everything"
- F — "The most creative person I know — everything they touch has their fingerprint on it"

Signal map: A=Problem-solver+execution+Analytical/Technical, B=Exploratory+scattered curious, C=Creative+inconsistency weakness, D=High execution+Strategic/Technical, E=Overthinking weakness+Analytical, F=Creative+strong identity

Input Format

```json
{
  "Q1": ["A"],
  "Q2": ["F", "A"],
  "Q3": ["B"],
  "Q4": ["A"],
  "Q5": ["D"]
}
```

Index 0 = primary (full weight), Index 1 = secondary (half weight)

Onboarding LLM Prompt

```
You are a behavioral classification engine for Zynth,
a gamified adaptive growth system.

You will receive answers to 5 situational scenario questions.
These questions are deliberately psychological and oblique —
they never mention learning, studying, or skills.
The user answered honestly because they didn't know what was being measured.

Your job: infer domain, profession, learning style, behavioral patterns,
and weakness signals from the behavioral choices each answer reveals.
Trust the psychology. Read between the lines.

OUTPUT MUST BE VALID JSON ONLY.
NO explanations. NO extra text. NO markdown code fences.

---

INPUT FORMAT:
Each question is an array of selected options.
  Index 0 = primary choice (full weight)
  Index 1 = secondary choice (half weight, if present)

---

USER ANSWERS:
{answers_json}
```

```
CLASSIFICATION RULES:

1. DOMAIN — assign ONE:
   ANALYTICAL  → logic, reasoning, data, precision, pattern-finding
   TECHNICAL   → building, systems, engineering, creating things
   CREATIVE    → design, expression, content, imagination, aesthetics
   SOCIAL      → communication, teaching, influence, connection
   STRATEGIC   → planning, business, decision-making, systems thinking

   Use signals from ALL 5 questions.
   A person who chooses "investor" in Q1 but reveals
   deep creative impulse in Q2–Q5 is CREATIVE, not STRATEGIC.
   Behavior always outweighs stated preference.

2. PROFESSION — specific aspirational title.
   Not a category — a real job title the user would feel proud of.
   Examples: "Software Developer", "Content Strategist",
   "Data Analyst", "UX Designer", "AI Systems Builder",
   "Brand Strategist", "Product Manager", "Educator"

3. LEARNING STYLE — assign ONE:
   Guided Learner      → needs structure, roadmaps, step-by-step
   Practical Learner   → learns by doing, jumps straight in
   Analytical Learner  → needs to understand WHY before HOW
   Exploratory Learner → self-directed, resists rigid paths

4. STATS — 6 total (4 core + 2 domain-specific):
   Core stats: focus, consistency, adaptability, execution
   Domain stats: pick 2 from the correct pool below

   Rules:
   - All 6 values between 4 and 10 (inclusive)
   - No two stats share the same value (every profile unique)
   - Total across all 6 stats: between 42 and 52
   - Domain stats returned as: { "name": "", "value": 0 }

   DOMAIN STAT POOLS:
   ANALYTICAL: Logic, Reasoning, Analysis, Precision, Insight,
               Memory, Pattern, Evaluation, Structure, Judgment
   TECHNICAL:  Coding, Systems, Debugging, Architecture, Automation,
               Efficiency, Scalability, Integration, Optimization,
Engineering
   CREATIVE:   Creativity, Imagination, Expression, Design, Storytelling,
               Originality, Aesthetics, Innovation, Visualization, Artistry
   SOCIAL:     Communication, Persuasion, Leadership, Empathy, Collaboration,
               Influence, Networking, Clarity, Confidence, Engagement
   STRATEGIC:  Planning, Strategy, Decision, Vision, Risk, Coordination,
               Prioritization, Management, Analysis, Execution
```

```
5. SKILLS — generate 3 to 5 starting skills:
   - lowercase_underscore format (display layer adds spaces)
   - Must reflect what this specific user is BEGINNING to develop
   - Derived from behavioral signals, NOT stated preferences
   - Examples: structured_thinking, rapid_prototyping,
     persuasive_writing, debugging_intuition, systems_mapping

6. BEHAVIOR SIGNATURE — compact lowercase_underscore label:
   Captures HOW this person operates, not what they want.
   Format: [speed]_[mode]_[trait] or [mode]_[trait]
   Examples: fast_builder_low_consistency,
   analytical_perfectionist, creative_explorer_scattered,
   quiet_executor, social_connector_reactive

7. WEAKNESS SIGNAL — the single biggest behavioral obstacle:
   Examples: poor_focus, inconsistency, overthinking,
   fear_of_failure, scattered_interests, laziness,
   decision_paralysis, external_dependency

8. DOMAIN CONFIDENCE — float 0.0 to 1.0
   How confident are you in the domain assignment?
   Mixed signals → lower confidence.

──────────────────────────────────────────────────

OUTPUT (strict — no extra fields):
{
  "profession": "",
  "domain": "",
  "domain_confidence": 0.0,
  "learning_style": "",
  "behavior_signature": "",
  "weakness_signal": "",
  "stats": {
    "focus": 0,
    "consistency": 0,
    "adaptability": 0,
    "execution": 0,
    "stat_5": { "name": "", "value": 0 },
    "stat_6": { "name": "", "value": 0 }
  },
  "skills": [
    { "name": "", "level": 1, "exp": 0 }
  ]
}
```

App Scaffolding (appended by app — never from LLM)

javascript

```
const scaffolding = {
  user_id: userId,
  level: 1,
  exp: 0,
  exp_to_next: 41,
  prestige: 0,
  prestige_exp_bonus: 1.0,
  title: 'Initiate',
  streak: 0,
  streak_best: 0,
  last_active: new Date().toISOString(),
  tokens: 3,
  xp_cards: 0,
  xp_card_log: [],
  swift_mode_unlocked: false,
  quest_history: [],
  behavior_log: [],
  hidden_skills: [],
  unlocks: [],
  memory: {
    milestones: [],
    stat_history: [],
    skill_history: [],
    pattern_summary: null,
    struggle_log: [],
    personal_bests: {
      highest_streak: 0,
      most_exp_in_day: 0,
      highest_skill_level: { name: '', level: 0 },
      fastest_level_up: { from_level: 0, to_level: 0, quests_taken: 0 }
    }
  },
  exploit_flags: [],
  current_adaptive_state: 'optimal',
  adaptive_state_updated_at: new Date().toISOString(),
  continuous_profile_answers: [],
  onboarding_stats: deepCopy(llmProfile.stats), // permanent floor
  schema_version: 1,
  created_at: new Date().toISOString(),
  updated_at: new Date().toISOString()
}
```

---

5. ENGINE 2: QUEST GENERATION ENGINE

Purpose

Generate one personalized quest per day. Adapts to profile, adaptive state, difficulty, and

quest type. LLM generates content. App resolves all logic.

Quest Types

Type	Trigger	Duration	EXP Multiplier	Tone
calibration	First quest only	2–5 min	1×	Welcoming, observational
daily	Every day	5–15 min	1×	Core habit loop
mission	Day 7 streak	20–45 min	3×	Epic, multi-step, fully in-app
challenge	Level 10+, 20% chance	15–30 min	5×	High stakes, deep
hidden	15% on Excellent grade	Surprise	2×	Mysterious
weakness	3 consecutive Fails	5–10 min	1×	Supportive, redemption
prestige	Level 100+, 10% chance	30–45 min	Massive	Legendary, elite

> **Non-negotiable rule: Every quest and mission is completed entirely inside the app in the text input in one sitting. The user never leaves the app to complete a task.**

Quest Type Resolver

```javascript
```

```
export function resolveQuestType(profile) {
  const { quest_history, streak, level } = profile

  // First quest ever → calibration
  if (!quest_history || quest_history.length === 0) return 'calibration'

  // Three consecutive Fails → weakness quest
  const lastThree = quest_history.slice(-3)
  if (lastThree.length === 3 && lastThree.every(q => q.grade === 'Fail')) {
    return 'weakness'
  }

  // Weekly mission — streak multiple of 7
  if (streak > 0 && streak % 7 === 0) {
    const lastMission = [...quest_history].reverse()
      .find(q => q.quest_type === 'mission')
    if (!lastMission || daysSince(lastMission.completed_at) >= 7) {
      return 'mission'
    }
  }

  // Challenge — level 10+, 20% chance
  if (level >= 10 && Math.random() < 0.20) return 'challenge'

  // Hidden — 15% chance after last Excellent
  const lastQuest = quest_history[quest_history.length - 1]
  if (lastQuest?.grade === 'Excellent' && Math.random() < 0.15) return 'hidden'

  // Prestige — level 100+, 10% chance
  if (level >= 100 && Math.random() < 0.10) return 'prestige'

  return 'daily'
}

export function calculateDifficulty(level) {
  return Math.max(1, Math.ceil(level / 5))
}
```

Quest Generation Prompt

```
You are the quest generation engine for Zynth,
a gamified adaptive growth system.

Generate ONE highly personalized quest for this user.

OUTPUT MUST BE VALID JSON ONLY.
NO explanations. NO extra text. NO markdown code fences.
```

```
──────────────────────────────────────────────────
PLAYER PROFILE (compressed):
{compressed_profile}

QUEST TYPE: {quest_type}
DIFFICULTY: {difficulty}
ADAPTIVE STATE: {adaptive_state}
──────────────────────────────────────────────────

ADAPTIVE INJECTION (follow this if present):
{prompt_injection}

──────────────────────────────────────────────────
IN-APP ONLY — NON-NEGOTIABLE RULE:
Every quest must be completable entirely inside this app,
in the text input box, in one sitting.

NEVER ask the user to:
- Build or code something outside the app
- Record audio or video
- Publish or share anything externally
- Open another app or browser
- Do research outside the app first

ALL tasks must be one of:
write · plan · explain · analyze · design · reflect ·
break down · describe · map · argue · evaluate · draft

REFRAME RULE — always apply this:
BAD:  "Build a to-do app and share the link."
GOOD: "Design a to-do app — components, data flow,
       and user interactions — entirely in writing."

BAD:  "Record a pitch for your idea."
GOOD: "Write your pitch word for word, exactly as you'd say it out loud."

BAD:  "Post content and share the link."
GOOD: "Write the full piece of content right here, ready to post as-is."
──────────────────────────────────────────────────

QUEST TYPE RULES:
calibration: First quest. Difficulty 1. 2–5 min. Welcoming, observational.
Completion = success.
daily: 5–15 min. Single prompt. Achievable but requires genuine thought.
mission: 20–45 min. Multi-step (3 steps max). Epic feel. Steps build on each
other.
challenge: 15–30 min. Single deep prompt. Genuinely difficult.
hidden: Mysterious tone. Unexpected angle. Feels like a secret level.
weakness: Difficulty-1. Addresses weakness_signal directly. Supportive,
```

```
short.
prestige: Level 100+. Legendary feel. Multi-step or ultra-deep.

LEARNING STYLE FORMATS:
Guided Learner      → step-by-step, clear criteria, no ambiguity
Practical Learner   → action-first, specific task, minimal theory
Analytical Learner  → explain the why, open-ended thinking required
Exploratory Learner → broad prompt, freedom to interpret

RULES:
- description must feel personally written for this user
- flavor_text: 1-sentence atmospheric line — like a quest board in an RPG
- skills_targeted: actual skill names from the user's profile
- Never use underscores in any display text
- For mission and prestige: populate the steps array
- For all other types: steps array is empty []

OUTPUT:
{
  "quest_title": "",
  "description": "",
  "type": "",
  "difficulty": 0,
  "estimated_time": "",
  "skills_targeted": [],
  "success_criteria": "",
  "hint": "",
  "flavor_text": "",
  "grading_type": "llm",
  "steps": [
    {
      "step_number": 1,
      "title": "",
      "prompt": "",
      "estimated_time": "",
      "success_criteria": ""
    }
  ]
}
```

Adaptive State → Quest Modification

State	Difficulty Mod	Quest Preference
optimal	0	Normal resolver
bored	+2	challenge, hidden
burned_out	-1	daily, short

State	Difficulty Mod	Quest Preference
stagnant	0	varied format
struggling	-2	weakness
returning	-1	daily, warm
peaking	+1	challenge, mission

6. ENGINE 3: GRADING ENGINE

Grade Scale

Grade	Base EXP	What it means
Excellent	15	Depth, genuine effort, surprising quality
Good	9	Meets criteria, solid work, minor gaps
Basic	6	Partial effort, lacks depth
Bad	3	Barely engages, minimal effort
Fail	0	Empty, off-topic, no genuine attempt

Streak EXP Multipliers

Streak	Multiplier
Day 1–6	1.0×
Day 7	1.5×
Day 14	1.75×
Day 30	2.0×
Day 60	2.5×
Day 100	3.0×

Grading Prompt

```
You are a mentor-evaluator inside Zynth,
a gamified adaptive growth system.
```

```
Evaluate the user's quest response and return structured feedback.
The LLM NEVER controls numbers. Return signals only.

OUTPUT MUST BE VALID JSON ONLY.
NO explanations. NO extra text. NO markdown code fences.

──────────────────────────────────────────────────

QUEST:
{quest_object}

USER RESPONSE:
{user_response}

PLAYER PROFILE (compressed):
{compressed_profile}
──────────────────────────────────────────────────

GRADING SCALE:
  Excellent → Deep understanding, genuine effort, surprising quality
  Good      → Meets success criteria, solid, minor gaps
  Basic     → Partially meets criteria, effort but lacks depth
  Bad       → Barely engages, significant misunderstanding
  Fail      → Empty, off-topic, or no genuine attempt

FEEDBACK RULES:
  - feedback: 2–3 sentences. Reference what they actually wrote. No
platitudes.
  - mentor_line: ONE punchy line matching their behavior_signature tone.
    fast_builder  → direct and charged
    analytical    → thoughtful and precise
    creative      → expressive and vivid
    quiet_executor → calm and affirming
  - strengths: specific things they did well in THIS response
  - improvements: specific push-further notes. Constructive only.

BEHAVIORAL SIGNALS (strings — never stat numbers):
  response_depth:       thorough | adequate | shallow | rushed
  domain_relevance:     high | medium | low
  difficulty_handled:   exceeded | met | struggled
  completion_quality:   strong | adequate | weak

SKILL PROGRESS: Only skills the user actually engaged in this response.
  exp_signal: high | medium | low (not a number)

HIDDEN QUEST TRIGGER: true only on Excellent, ~15% of the time.
SKILL EVOLUTION SIGNAL: skill name if 10th meaningful targeting. Else null.

OUTPUT:
{
```

```
  "grade": "Excellent|Good|Basic|Bad|Fail",
  "feedback": "",
  "mentor_line": "",
  "strengths": [],
  "improvements": [],
  "behavioral_signals": {
    "response_depth": "",
    "domain_relevance": "",
    "difficulty_handled": "",
    "completion_quality": ""
  },
  "skill_progress": [
    { "name": "", "exp_signal": "high|medium|low" }
  ],
  "hidden_quest_triggered": false,
  "skill_evolution_signal": null
}
```

EXP Calculation (App-Side)

javascript

```
const GRADE_BASE_EXP = {
  Excellent: 15, Good: 9, Basic: 6, Bad: 3, Fail: 0
}

export function calculateStreakMultiplier(streak) {
  if (streak >= 100) return 3.0
  if (streak >= 60)  return 2.5
  if (streak >= 30)  return 2.0
  if (streak >= 14)  return 1.75
  if (streak >= 7)   return 1.5
  return 1.0
}

export function calculateFinalEXP(grade, streak, prestigeBonus = 1.0) {
  const base = GRADE_BASE_EXP[grade] || 0
  return Math.round(base * calculateStreakMultiplier(streak) * prestigeBonus)
}

export function calculateExpToNext(level) {
  return Math.round(30 * level * (1 + Math.log(level + 1)))
}

export function checkLevelUp(currentLevel, currentExp, addedExp) {
  let level = currentLevel
  let exp = currentExp + addedExp
  let didLevelUp = false

  while (true) {
    const required = calculateExpToNext(level)
    if (exp >= required) {
      exp -= required
      level++
      didLevelUp = true
    } else break
  }

  return {
    didLevelUp,
    newLevel: level,
    newExp: exp,
    newExpToNext: calculateExpToNext(level),
    newTitle: didLevelUp ? getTitleForLevel(level) : null
  }
}
```

Post-Grade 14-Step Pipeline (App Logic)

```
Step 1:  Parse grade from LLM result
```

```
Step 2:  Run anti-exploit check — apply EXP penalty if soft_block (×0.3)
Step 3:  calculateFinalEXP(grade, streak, prestige_bonus)
Step 4:  checkLevelUp(level, exp, earnedExp)
Step 5:  updateStreak: if grade !== 'Fail' AND quest completed same day →
streak++
         updateTokens: if streak hits milestone AND tokens < 5 → tokens++
Step 6:  Append to behavior_log (cap at 50, drop oldest)
Step 7:  runStatEngine(updatedBehaviorLog)
Step 8:  Apply stat deltas (floor = onboarding_stats values)
Step 9:  updateSkillExp(skills, skill_progress from LLM)
Step 10: checkSkillEvolution(skill_evolution_signal from LLM)
Step 11: checkAchievements(updatedProfile)
Step 12: awardXPCards(events triggered)
Step 13: writeMemoryEvents(events)
Step 14: checkProfessionEvolution(profile)
         checkWeaknessOvercome(profile)
         → Write ALL updates to Supabase in ONE atomic operation
         → Queue event screens for UI
```

---

7. ENGINE 4: BEHAVIORAL STAT ENGINE

Philosophy

```
Stats are not rewards. They are reflections.
The system watches patterns over time, not individual events.
You cannot grind stats. You have to actually change your behavior.
```

Observation Windows

```
Short window: last 5 quests  → catches recent behavior shifts
Long window:  last 20 quests → catches deep habits
```

The 6 Stats — Full Behavior Map

FOCUS

Behavior	Effect	Window
response_depth 'thorough' (3+ of 5)	+1	Short
response_depth 'rushed' (3+ of 5)	-1	Short
Excellent/Good pattern (4+ of 5)	+1	Short
Bad/Fail pattern (3+ of 5)	-1	Short

CONSISTENCY

Behavior	Effect	Window
Streak milestones (7,14,30,60,100)	+2 to +5	Event
Completes same day (4+ of 5)	+1	Short
Skips without token (2+ of 5)	-1	Short
Skips without token (3+ of 5)	-2	Short
Returns after break + completes	+1	Recovery

ADAPTABILITY

Behavior	Effect	Window
difficulty_handled 'exceeded' (2+ of 5)	+1	Short
difficulty_handled 'struggled' (3+ of 5)	-1	Short
Consistent across different quest types	+1	Long

EXECUTION

Behavior	Effect	Window
4+ completions in last 5	+1	Short
3 consecutive Fails	-1	Short
completion_quality 'strong' (3+ of 5)	+1	Short

Weakness Stat Modifier

```javascript
```

```
const WEAKNESS_MAP = {
  poor_focus: 'focus',
  inconsistency: 'consistency',
  overthinking: 'adaptability',
  fear_of_failure: 'execution',
  laziness: 'execution',
  scattered_interests: 'adaptability',
  decision_paralysis: 'adaptability',
  external_dependency: 'consistency'
}

// If weakness stat would improve, reduce delta by 25%
// Breaking your pattern is harder than building a new one
if (weaknessStat && statDeltas[weaknessStat] > 0) {
  statDeltas[weaknessStat] = Math.floor(statDeltas[weaknessStat] * 0.75)
}
```

Stat Floor Rule

javascript

```
// Stats never go below onboarding_stats values
// onboarding_stats is a frozen snapshot taken at profile creation
const newValue = Math.max(
  profile.onboarding_stats[statName],
  profile.stats[statName] + delta
)
```

Skill EXP System

javascript

```
const EXP_SIGNALS = { high: 20, medium: 10, low: 5 }

export function calculateSkillExpToNext(skillLevel) {
  return Math.round(3 * skillLevel * (1 + Math.log(skillLevel + 1)))
}

// Skill tiers
// Level 1-3:   Tier 1 — Raw
// Level 4-9:   Tier 2 — Developing
// Level 10-24: Tier 3 — Proficient
// Level 25-49: Tier 4 — Advanced
// Level 50-99: Tier 5 — Expert
// Level 100-199: Tier 6 — Mastery
// Level 200+:  Tier 7 — Transcendent
```

Weakness Overcome Detection

```javascript
const OVERCOME_THRESHOLD = 10 // positive signals in last 14 log entries

export function checkWeaknessOvercome(profile) {
  const weaknessStat = WEAKNESS_MAP[profile.weakness_signal]
  if (!weaknessStat) return null

  const last14 = profile.behavior_log.slice(-14)
  const positiveSignals = last14.filter(e => {
    if (weaknessStat === 'focus') return e.response_depth === 'thorough'
    if (weaknessStat === 'consistency') return e.completedSameDay
    if (weaknessStat === 'adaptability') return e.difficulty_handled === 'excee
    if (weaknessStat === 'execution') return e.completion_quality === 'strong'
    return false
  }).length

  if (positiveSignals >= OVERCOME_THRESHOLD) {
    return {
      overcome: true,
      stat: weaknessStat,
      bonus: 3 // permanent +3 to that stat
    }
  }
  return null
}
```

---

8. ENGINE 5: PROGRESSION ENGINE

Level Titles

Level Range	Title
1–4	Initiate
5–9	Apprentice
10–19	Journeyman
20–34	Forgehand
35–49	Craftsman
50–74	Artisan
75–99	Specialist
100–149	Expert

Level Range	Title
150–199	Master
200–299	Grandmaster
300–499	Legendary
500–999	Mythic
1000+	LLM-generated unique legendary title

Level Unlocks

Level	Unlock
5	Swift Mode
10	Challenge Quests
20	Memory Context injected into quests
100	Prestige available
100	Prestige Quests

Prestige System

```
Prestige = voluntary reset at Level 100
Character level resets to 1
Stats soft-reset to (onboarding_stats + 2 per prestige)
Skills retain their evolution names but reset to Level 1
prestige_exp_bonus += 0.25 per prestige (permanent EXP multiplier)
prestige counter increments
New prestige badge visible on profile
```

---

9. ENGINE 6: HIDDEN SKILL ENGINE

Philosophy

```
Everyone has abilities they haven't named yet.
Zynth watches for behavioral patterns the user
doesn't know they're demonstrating.
The discovery moment is designed to feel like revelation.
```

Hidden Skill Pool

Skill	Domain	Trigger Condition
Systems Thinking	ANALYTICAL	References interconnected systems in 5+ responses
Discipline	Any	30-day streak achieved
Pattern Recognition	ANALYTICAL	Finds non-obvious patterns in 3+ analysis quests
Mentorship	SOCIAL	Explains things to others in 4+ responses
Leadership	SOCIAL	Takes initiative / proposes direction in 5+ responses
Resilience	Any	Excellent grade after 3+ consecutive Fails
Creative Synthesis	CREATIVE	Combines unrelated concepts in 3+ responses
Product Thinking	STRATEGIC	Balances user, business, tech in 3+ responses
Operations	STRATEGIC	Plans logistics/workflows in 4+ responses
Deep Focus	Any	5+ 'thorough' response depths in a row

Discovery Flow



Discovery Message Prompt

```
A user just discovered a hidden skill in Zynth.
Write a 2–3 sentence discovery message that feels like
uncovering something that was always there.
Tone: revelatory, personal, slightly mysterious.
```

```
User profession: {profession}
User behavior signature: {behavior_signature}
Discovered skill: {skill_name}

Return ONLY the message text. Nothing else.
```

Profile UI Representation

```
HIDDEN SKILLS

  DISCOVERED
  ├── Systems Thinking  Lv 1  [teal]
  └── Discipline        Lv 3  [teal]

  UNKNOWN
  ├── ???
  ├── ???
  └── ???

The ??? slots are always visible from day one.
The user knows something is there.
That ambient tension makes the reveal land harder.
```

---

10. ENGINE 7: ANTI-EXPLOITATION ENGINE

Philosophy

```
The system must be harder to fool than to use honestly.
Detection is invisible to legitimate users.
Never tell users exactly what triggered a flag.
```

Detection Patterns

Pattern	Threshold	Severity
RAPID_SUBMISSION < 30s	Immediate	soft_block
RAPID_SUBMISSION < 60s	Immediate	warn
LOW_EFFORT (< 50 chars)	Immediate	warn
DUPLICATE_RESPONSE (>85% similar to last 5)	Immediate	hard_block
LOW_EFFORT_PATTERN (3+ weak in 7 quests)	7 quests	soft_block

Pattern	Threshold	Severity
STREAK_ABUSE (Bad/Fail 5+ days + <90s submit)	7 quests	hard_block

Consequence Ladder

Level	Action	User Sees
warn	Log only, no penalty	Nothing
soft_block	EXP × 0.3, strict grader injected	Nothing
hard_block	Quest rejected	Vague "try again" message
ban_review	3+ hard_blocks in 7 days	Nothing (flagged internally)

Grader Injection (appended silently on soft_block)

```
IMPORTANT: This response has been flagged for low engagement signals.
Apply strict grading criteria.
Only award Excellent or Good if the response demonstrates genuine
personal engagement, specific examples, and depth beyond the obvious.
A response that barely meets surface criteria should receive Basic or Bad.
```

---

11. ENGINE 8: CONTEXT COMPRESSION ENGINE

Token Budgets

Call Type	Max Input Tokens
quest_generation	800
grading	600
hidden_skill_detection	400
skill_evolution_naming	200
behavior_summary	300
daily_profile_question	150

Compression Map

Raw Data	Compressed To
Full behavior_log (50 events)	2-sentence summary (cached every 10 quests)
Full quest_history	Last 5 grades + dominant grade
Full skills array	"skill_name (LvN)" comma-separated
Stats with metadata	name:value pairs only
Hidden skills	Discovered names only
Memory object	Max 5 key facts (Level 20+ only)
Continuous profile answers	Last 10 most distinctive answers

```javascript
```

```
export function buildCompressedContext(profile, purpose) {
  const base = {
    profession: profile.profession,
    domain: profile.domain,
    behavior_signature: profile.behavior_signature,
    weakness_signal: profile.weakness_signal,
    level: profile.level,
    stats: compressStats(profile.stats)
  }

  if (purpose === 'quest_generation') {
    return {
      ...base,
      skills: compressSkills(profile.skills),
      hidden_skills_discovered: profile.hidden_skills
        .filter(s => s.discovered).map(s => s.name),
      recent_performance: profile.quest_history.slice(-5).map(q => q.grade),
      memory_context: profile.level >= 20 ? buildMemoryContext(profile) : null,
      continuous_profile_insights: buildProfileInsights(profile)
    }
  }

  if (purpose === 'grading') {
    return {
      ...base,
      skills: compressSkills(profile.skills),
      recent_grades: profile.quest_history.slice(-5).map(q => q.grade)
    }
  }
}
```

---

12. ENGINE 9: MEMORY ENGINE

Memory Schema

javascript

```
const MEMORY_SCHEMA = {
  milestones: [
    { type: 'first_excellent', quest_title: '', at_level: 0, date: '' },
    { type: 'first_skill_evolution', from_skill: '', to_skill: '', at_level: 0,
    { type: 'longest_streak', streak_days: 0, date: '' },
    { type: 'first_hidden_skill', skill: '', at_level: 0, date: '' },
    { type: 'profession_evolution', from: '', to: '', at_level: 0, date: '' },
    { type: 'prestige', prestige_number: 0, date: '' },
    { type: 'weakness_overcome', weakness: '', stat: '', at_level: 0, date: ''
  ],
  stat_history: [
    { at_level: 10, stats: {}, date: '' }
    // snapshot every 10 levels
  ],
  skill_history: [
    { skill_name: '', evolutions: [], current_level: 0 }
  ],
  pattern_summary: null, // 2-sentence LLM summary, updated every 20 quests
  struggle_log: [
    { quest_type: '', domain: '', fail_count: 0, eventually_passed: false }
  ],
  personal_bests: {
    highest_streak: 0,
    most_exp_in_day: 0,
    highest_skill_level: { name: '', level: 0 },
    fastest_level_up: { from_level: 0, to_level: 0, quests_taken: 0 }
  }
}
```

Memory Write Events

Event	What Gets Written
first_excellent	Quest title + level (once ever)
skill_evolution	Old name, new name, level
level_milestone	Full stat snapshot every 10 levels
streak_broken	Save to personal_bests first
quest_failed	Add to struggle_log
hidden_skill_discovered	Skill name, level at discovery
profession_evolution	From, to, triggering skills
prestige	Prestige number

Event	What Gets Written
weakness_overcome	Weakness signal, stat boosted, level

13. ENGINE 10: ECONOMY ENGINE

Two Currencies

```
◈  SKIP TOKENS — earned through consistency
⬡  XP CARDS    — earned through performance
```

Skip Token Rules

Rule	Value
Starting balance	3
Max held	5
Earn rate	1 per 7-day streak milestone
Expiry	Never
Cost: skip quest	1 token
Cost: Swift Mode early (before Lv5)	2 tokens

XP Card Rules

Rule	Value
Starting balance	0
Max held	20
Expiry	30 days from earn date
Cost: retest	2 cards
Cost: reduce difficulty	1 card
Cost: hint upgrade	1 card
Cost: bonus quest	3 cards

XP Card Earn Events

Event	Cards
Grade: Excellent	1
First Excellent ever	3
Mission complete	2
Challenge Quest complete	3
Hidden skill discovered	5
7-day streak	1
30-day streak	3
100-day streak	10
Skill evolution	2
Level milestone (every 5 levels)	1
Prestige	15
Weakness overcome	5
Achievement unlocked	1

14. ENGINE 11: ACHIEVEMENT ENGINE

Philosophy

```
Achievements are not participation trophies.
They are discovered, not unlocked.
The best ones the user will never see coming.
```

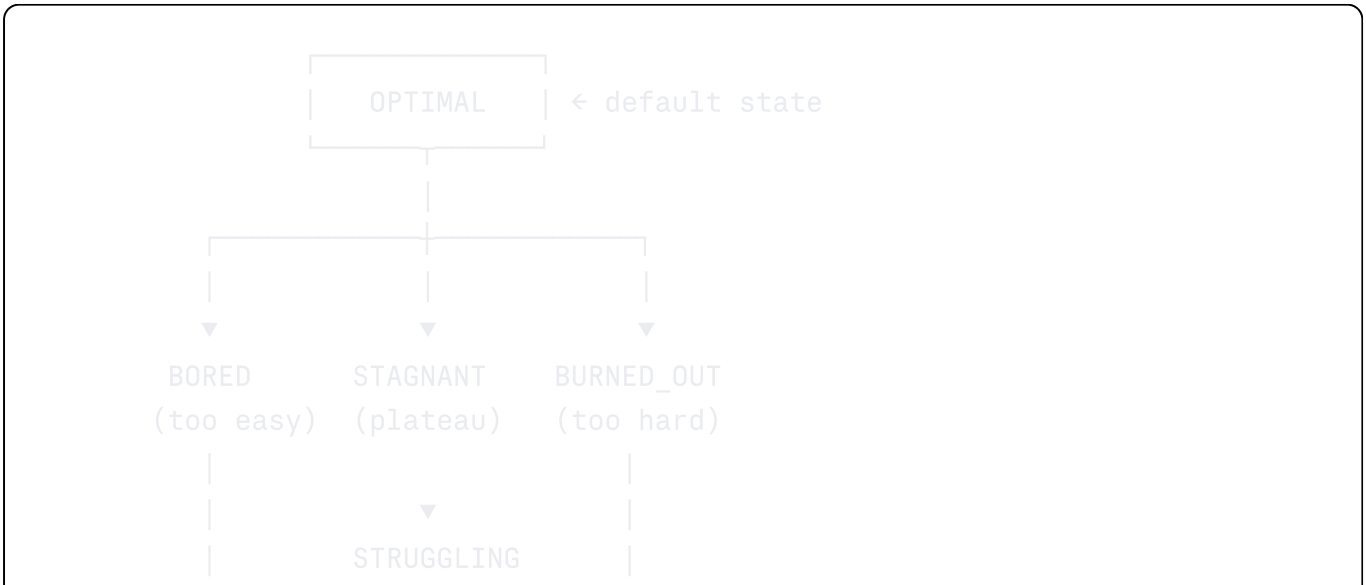
Achievement Library

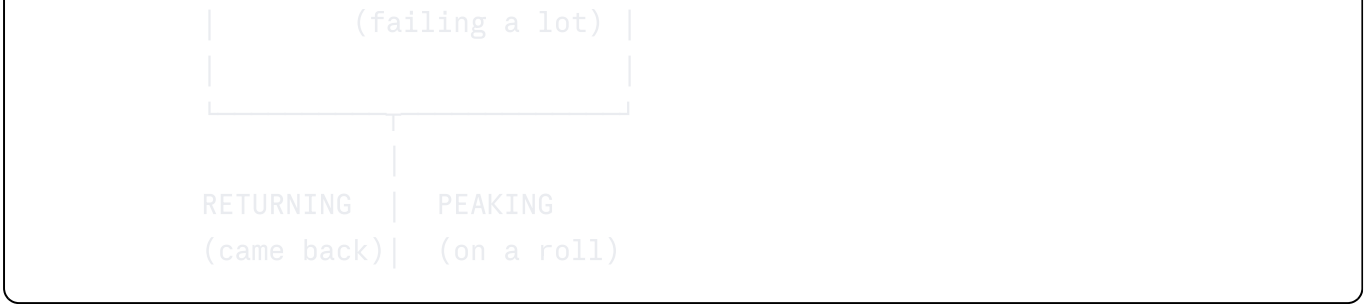
ID	Name	Condition	Rarity	XP Cards	Visible
first_flame	First Flame	3-day streak	Common	0	Yes
week_forged	Week Marked	7-day streak	Common	1	Yes
century_streak	Century	100-day streak	Legendary	10	Yes

ID	Name	Condition	Rarity	XP Cards	Visible
first_excellent	Sharp	First Excellent	Common	1	Yes
flawless_week	Flawless Week	Excellent/Good all 7 days	Rare	3	Hidden
comeback_king	Comeback	Excellent right after Fail	Rare	2	Hidden
apprentice	Apprentice	Reach Level 5	Common	0	Yes
triple_digits	Triple Digits	Reach Level 100	Legendary	15	Yes
prestige_one	Reborn	First Prestige	Legendary	15	Yes
first_evolution	Metamorphosis	First skill evolution	Uncommon	2	Yes
master_tier	Mastery	Tier 6 on any skill	Legendary	10	Yes
shadow_talent	Shadow Talent	First hidden skill	Rare	5	Slot visible
triple_shadow	???	3 hidden skills	Legendary	8	Fully hidden
silent_discipline	???	30 quests without Swift Mode	Rare	4	Fully hidden
the_patient	???	Fail 3×, then Excellent 4th	Legendary	6	Fully hidden

15. ENGINE 12: ADAPTIVE INTELLIGENCE ENGINE

State Machine





State Detection

javascript

```
export function detectUserState(profile) {
  const recentLog    = profile.behavior_log.slice(-10)
  const recentGrades = profile.quest_history.slice(-7).map(q => q.grade)
  const daysSinceActive = daysSince(profile.last_active)

  const scores = {
    bored: 0, burned_out: 0, stagnant: 0,
    struggling: 0, returning: 0, peaking: 0, optimal: 0
  }

  // BORED
  const excellentTrail = getTrailingCount(recentGrades, 'Excellent')
  if (excellentTrail >= 5) scores.bored += 3
  const rushed = recentLog.filter(e => e.response_depth === 'rushed').length
  if (rushed >= 3) scores.bored += 2

  // BURNED_OUT
  const failCount = recentGrades.filter(g => g === 'Fail').length
  if (failCount >= 3) scores.burned_out += 3
  if (daysSinceActive >= 2 && daysSinceActive < 7) scores.burned_out += 2

  // STAGNANT
  const basicCount = recentGrades.filter(g => g === 'Basic').length
  if (basicCount >= 4) scores.stagnant += 2

  // STRUGGLING
  if (failCount >= 4) scores.struggling += 4
  const badCount = recentGrades.filter(g => g === 'Bad').length
  if (badCount + failCount >= 5) scores.struggling += 3

  // RETURNING
  if (daysSinceActive >= 3) scores.returning += 5

  // PEAKING
  if (excellentTrail >= 3 && profile.streak >= 7) scores.peaking += 4

  scores.optimal += 1

  const dominant = Object.entries(scores).sort(([,a],[,b]) => b - a)[0]
  return { state: dominant[0], confidence: dominant[1] }
}
```

State Prompt Injections

```
BORED:
  "This user is showing boredom signals — too many easy Excellents.
   Generate a quest that genuinely challenges them.
   Push the difficulty. Make them think harder than usual."
```

```
BURNED_OUT:
  "This user is showing burnout signals — recent failures, reduced
engagement.
   Generate a shorter, more achievable quest. Tone: supportive.
   Goal is to rebuild momentum, not challenge further."

STAGNANT:
  "This user has plateaued. Generate a quest that approaches their
   skills from a completely different angle. Break the format pattern."

STRUGGLING:
  "This user is struggling significantly. Be gentle.
   Generate the simplest possible version of a quest in their domain.
   This quest should feel winnable."

RETURNING:
  "This user is returning after an absence.
   Generate a re-entry quest that feels like a warm welcome back.
   Make them feel like the system remembers them. Keep it short."

PEAKING:
  "This user is on a peak run — excellent grades, strong streak.
   Match their energy. Raise the stakes slightly.
   Make this feel like a chapter boss fight."
```

---

16. ENGINE 13: CONTINUOUS PROFILING ENGINE

Philosophy

```
Onboarding gives Zynth a starting point.
Engine 13 never stops refining it.

Every day, one oblique psychological question is asked.
The user thinks it's a fun personality prompt.
Zynth is building the most accurate behavioral model
it has ever seen of this person.

By day 365, the profile is unrecognizable from day 1.
Every quest feels more personal than the last.
```

Daily Question Flow

```
User opens app
      |
      ▼
```

```
Energy Check-in (HIGH / MID / LOW)
[feeds Adaptive Intelligence Engine]
        |
        ▼
Daily Profiling Question
[one question, one tap answer]
        |
        ▼
Answer stored in continuous_profile_answers[]
        |
        ▼
Engine 8 compresses into profile insights
        |
        ▼
Quest generation uses enriched profile
```

Question Rules

```
- One question per day
- Always oblique — NEVER mention learning, skills, studying, growth
- Options must be mutually exclusive and revealing
- No option should feel obviously "better" than others
- User can skip (single tap) — skip itself is logged as a signal
- Questions rotate through categories, tracking what's already known
- After 30 questions: re-ask calibration questions to detect drift
```

Question Categories and Rotation

Category	What it mines	Frequency
Time & environment	When/where they operate best	Weekly
Mood & energy	What conditions produce best work	Weekly
Social context	Solo vs collaborative preference	Bi-weekly
Risk tolerance	Conservative vs exploratory	Bi-weekly
Emotional regulation	Response to failure, pressure	Bi-weekly
Ambition signals	Short vs long term thinking	Monthly
Identity shifts	How they see themselves changing	Monthly
Preference drift	Re-check established assumptions	Every 30 days

Example Daily Questions

Time & Environment: *"Pick the version of your day where you'd get the most done:"*

- A — Starting at 5am before the world wakes up
- B — Deep afternoon, after everything's been figured out for the day
- C — Late at night when no one can interrupt
- D — Honestly? I'm equally useless at all of them

Risk & Curiosity: *"You find an unmarked door in a building you know well. Do you open it?"*

- A — Immediately. Obviously.
- B — Only if I can figure out what's behind it first
- C — I note that it's there and move on
- D — I'd probably mention it to someone else and let them decide

Emotional Regulation: *"Someone cancels plans on you last minute. Your honest first reaction:"*

- A — Relief, honestly
- B — Annoyed — I had organized around this
- C — I immediately start filling the time with something useful
- D — I check in to make sure they're okay first

Ambition Signals: *"A stranger offers you: take $5,000 now, or wait 2 years for $50,000. You:"*

- A — Take the $5,000. The future is uncertain.
- B — Wait for the $50,000. The math is obvious.
- C — Ask a hundred questions before deciding anything
- D — Depends entirely on what I need right now

Identity: *"Which version of 'successful' actually sounds appealing to you:"*

- A — Known by a small group as genuinely exceptional
- B — Widely recognized for something that matters
- C — Financially free to do whatever I want
- D — I don't think about it that way — I just want to feel good about my days

Daily Question Generation Prompt

```
Generate one oblique psychological scenario question for Zynth's
daily profiling system.

RULES:
- NEVER mention learning, studying, skills, growth, productivity,
  goals, self-improvement, or anything related to the app's purpose
- The question must feel like a fun personality curiosity
- Options must be mutually exclusive and genuinely revealing
- No option should feel obviously "better" than others
- The question fits this category: {category}
```

```
- Avoid repeating themes from these recent questions: {recent_themes}

OUTPUT FORMAT (JSON only, no markdown):
{
  "question": "",
  "options": [
    { "key": "A", "label": "" },
    { "key": "B", "label": "" },
    { "key": "C", "label": "" },
    { "key": "D", "label": "" }
  ],
  "category": "",
  "signals_mined": ""
}
```

---

17. LEADERBOARD SYSTEM

Philosophy

```
Competition makes growth personal.
The leaderboard doesn't just show who's winning —
it gives everyone a reason to keep climbing.
Every position matters. Not just the top 10.
```

Leaderboard Tiers

Leaderboard	Scope	Basis
Local	Same city/region	Cumulative EXP
State	Same state/province	Cumulative EXP
National	Same country	Cumulative EXP
Global	Worldwide	Cumulative EXP

Each leaderboard has two views:

- **By Skill Category** — compete only against users in same domain (TECHNICAL, CREATIVE, etc.)
- **All Skills** — compete against everyone regardless of domain

Leaderboard → Profession Evolution Link

```
Climbing the cumulative skill leaderboard can trigger profession evolution.
```

```
Top 25% in your skill category  → eligible for Senior tier profession
Top 10% in your skill category  → eligible for Lead tier profession
Top 5% globally in your category → eligible for Principal/Director tier

Examples:
  Software Developer + Top 10% TECHNICAL → Senior Software Developer
  Content Creator + Top 25% CREATIVE     → Senior Content Creator
  Designer + Top 5% globally             → Creative Director

This is ONE path to profession evolution.
Hidden skill discovery remains the other path.
Neither gates the other.
```

Leaderboard Calculation

javascript

```
// Calculated server-side via Supabase Edge Function
// Runs once per day at midnight UTC

export async function calculateLeaderboards() {
  // 1. Fetch all profiles with cumulative_exp > 0
  // 2. Group by domain (for skill category boards)
  // 3. Rank by cumulative_exp within each group
  // 4. Rank by cumulative_exp across all groups (global)
  // 5. Determine local/state/national scope from user location (optional — can
  // 6. Cache results in leaderboard_cache table
  // 7. Update profession evolution eligibility flags
}
```

Privacy Rules

```
- Display: first name + last initial only ("Marcus T.")
- No email, no full name, no location below national level
- User can opt out of leaderboard entirely in Settings
- Opt-out does not affect any other mechanic
```

---

18. THE ADDICTION LAYER

The Four Hooks

```
1. VARIABLE REWARDS
   Hidden quests (15% chance on Excellent)
   Surprise EXP multipliers on streaks
   Random hidden skill unlock events
   ??? slots visible but unrevealed
```

```
   Daily profiling question reveals something new about you

2. IMMEDIATE FEEDBACK
   Grade card flips instantly after submit
   EXP bar animates in real time
   Stats pulse on change
   Mentor line hits before feedback paragraph
   Character visually reacts to grade

3. PROGRESS ALWAYS VISIBLE
   Level + EXP bar always on screen
   Streak flame always visible
   Skills show individual level bars
   Mark map shows how far you've come

4. SUNK COST + IDENTITY
   Behavior signature updates as you grow
   Skills evolve with unique names you earned
   Titles change and feel earned
   Character reacts to your absence
   The profile is YOU — not a game avatar
```

Character Reaction System

```
Character visual state changes based on engagement:

ACTIVE (completed quest today): Character glows, full color, particles
STREAK (7+ days): Amber aura, stands taller
PEAK (3+ Excellents): Golden particles, animated
ABSENT (3+ days): Character dims, slightly desaturated
LONG ABSENT (7+ days): Character faded, sitting posture
JUST LEVELED UP: Scale animation, burst of light on open
WEAKNESS OVERCOME: Aura color changes to domain color permanently
```

Immediate Reward Moments

Moment	Animation
Quest submitted	"Evaluating..." pulse 1.5s
Grade reveal	Card flip + color burst
EXP earned	Counter animates up
Level up	Number explodes large
Skill evolution	Old name strikes, new fades in
Hidden skill	Full screen typewriter reveal

Moment	Animation
Streak milestone	Flame grows, multiplier badge drops
Achievement	Badge materializes with particle burst
Mark completed on roadmap	Node fills, path segment illuminates

19. DATABASE SCHEMA

```sql
```

```sql
-- ─── PROFILES ──────────────────────────────────────────────────────
CREATE TABLE profiles (
  id                          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id                     UUID REFERENCES auth.users(id) UNIQUE,

  -- Identity
  profession                  TEXT NOT NULL,
  domain                      TEXT NOT NULL,
  domain_confidence           FLOAT,
  learning_style              TEXT NOT NULL,
  behavior_signature          TEXT NOT NULL,
  weakness_signal             TEXT,
  title                       TEXT DEFAULT 'Initiate',
  schema_version              INTEGER DEFAULT 1,

  -- Progression
  level                       INTEGER DEFAULT 1,
  exp                         INTEGER DEFAULT 0,
  exp_to_next                 INTEGER DEFAULT 41,
  cumulative_exp              INTEGER DEFAULT 0,
  prestige                    INTEGER DEFAULT 0,
  prestige_exp_bonus          FLOAT DEFAULT 1.0,

  -- Engagement
  streak                      INTEGER DEFAULT 0,
  streak_best                 INTEGER DEFAULT 0,
  last_active                 TIMESTAMPTZ DEFAULT NOW(),
  tokens                      INTEGER DEFAULT 3,
  xp_cards                    INTEGER DEFAULT 0,
  xp_card_log                 JSONB DEFAULT '[]',
  swift_mode_unlocked         BOOLEAN DEFAULT FALSE,
  current_adaptive_state      TEXT DEFAULT 'optimal',
  adaptive_state_updated_at   TIMESTAMPTZ DEFAULT NOW(),

  -- JSON blobs
  stats                       JSONB NOT NULL,
  onboarding_stats            JSONB NOT NULL,
  skills                      JSONB DEFAULT '[]',
  hidden_skills               JSONB DEFAULT '[]',
  behavior_log                JSONB DEFAULT '[]',
  behavior_summary_cache      TEXT,
  behavior_summary_quest_count INTEGER DEFAULT 0,
  unlocks                     JSONB DEFAULT '[]',
  quest_history               JSONB DEFAULT '[]',
  exploit_flags               JSONB DEFAULT '[]',
  memory                      JSONB DEFAULT '{}',
  continuous_profile_answers  JSONB DEFAULT '[]',
```

```
  -- Leaderboard
  leaderboard_opt_out           BOOLEAN DEFAULT FALSE,
  display_name                  TEXT,

  -- Character
  character_name                TEXT,
  character_gender              TEXT,
  character_avatar              TEXT,

  created_at                    TIMESTAMPTZ DEFAULT NOW(),
  updated_at                    TIMESTAMPTZ DEFAULT NOW()
);

-- ─── QUESTS ──────────────────────────────────────────────────────
CREATE TABLE quests (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id           UUID REFERENCES auth.users(id),
  quest_data        JSONB NOT NULL,
  quest_type        TEXT NOT NULL,
  difficulty        INTEGER NOT NULL,
  status            TEXT DEFAULT 'active',
  is_multi_step     BOOLEAN DEFAULT FALSE,
  steps             JSONB DEFAULT '[]',
  step_responses    JSONB DEFAULT '[]',
  current_step      INTEGER DEFAULT 0,
  response          TEXT,
  grade             TEXT,
  points            INTEGER,
  exp_earned        INTEGER,
  behavioral_signals JSONB,
  swift_mode_used   BOOLEAN DEFAULT FALSE,
  retest_count      INTEGER DEFAULT 0,
  created_at        TIMESTAMPTZ DEFAULT NOW(),
  completed_at      TIMESTAMPTZ
);

-- ─── EVENTS ──────────────────────────────────────────────────────
CREATE TABLE events (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID REFERENCES auth.users(id),
  event_type  TEXT NOT NULL,
  payload     JSONB,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

-- ─── EXPLOIT LOG ─────────────────────────────────────────────────
CREATE TABLE exploit_log (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID REFERENCES auth.users(id),
```

```
  flag_type    TEXT NOT NULL,
  severity     TEXT NOT NULL,
  payload      JSONB,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);

-- ─── HIDDEN SKILL SIGNALS ──────────────────────────────────────
CREATE TABLE hidden_skill_signals (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id      UUID REFERENCES auth.users(id),
  signal_type  TEXT NOT NULL,
  signal_data  JSONB,
  created_at   TIMESTAMPTZ DEFAULT NOW()
);

-- ─── PROFESSION EVOLUTIONS ─────────────────────────────────────
CREATE TABLE profession_evolutions (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id           UUID REFERENCES auth.users(id),
  from_profession   TEXT,
  to_profession     TEXT,
  triggering_skills JSONB,
  trigger_type      TEXT, -- 'hidden_skills' or 'leaderboard'
  evolved_at        TIMESTAMPTZ DEFAULT NOW()
);

-- ─── LEADERBOARD CACHE ─────────────────────────────────────────
CREATE TABLE leaderboard_cache (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  board_type   TEXT NOT NULL, -- 'global_all', 'global_technical', etc.
  user_id      UUID REFERENCES auth.users(id),
  display_name TEXT,
  rank         INTEGER,
  score        INTEGER,
  domain       TEXT,
  level        INTEGER,
  streak       INTEGER,
  updated_at   TIMESTAMPTZ DEFAULT NOW()
);

-- ─── DAILY PROFILE QUESTIONS ───────────────────────────────────
CREATE TABLE daily_profile_questions (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id      UUID REFERENCES auth.users(id),
  question     TEXT NOT NULL,
  options      JSONB NOT NULL,
  category     TEXT NOT NULL,
  answer_key   TEXT,
  skipped      BOOLEAN DEFAULT FALSE,
```

```
  asked_at    TIMESTAMPTZ DEFAULT NOW(),
  answered_at TIMESTAMPTZ
);

-- ── INDEXES ──────────────────────────────────────────────────
CREATE INDEX idx_quests_user_id       ON quests(user_id);
CREATE INDEX idx_quests_status        ON quests(status);
CREATE INDEX idx_events_user_id       ON events(user_id);
CREATE INDEX idx_events_type          ON events(event_type);
CREATE INDEX idx_hss_user_id          ON hidden_skill_signals(user_id);
CREATE INDEX idx_leaderboard_type     ON leaderboard_cache(board_type);
CREATE INDEX idx_leaderboard_rank     ON leaderboard_cache(rank);
CREATE INDEX idx_dpq_user_id          ON daily_profile_questions(user_id);
CREATE INDEX idx_profiles_cumexp      ON profiles(cumulative_exp DESC);

-- ── ROW LEVEL SECURITY ──────────────────────────────────────
ALTER TABLE profiles               ENABLE ROW LEVEL SECURITY;
ALTER TABLE quests                 ENABLE ROW LEVEL SECURITY;
ALTER TABLE events                 ENABLE ROW LEVEL SECURITY;
ALTER TABLE exploit_log            ENABLE ROW LEVEL SECURITY;
ALTER TABLE hidden_skill_signals   ENABLE ROW LEVEL SECURITY;
ALTER TABLE profession_evolutions  ENABLE ROW LEVEL SECURITY;
ALTER TABLE daily_profile_questions ENABLE ROW LEVEL SECURITY;
ALTER TABLE leaderboard_cache      ENABLE ROW LEVEL SECURITY;

CREATE POLICY "own_profile"    ON profiles               FOR ALL USING (auth.ui
CREATE POLICY "own_quests"     ON quests                 FOR ALL USING (auth.ui
CREATE POLICY "own_events"     ON events                 FOR ALL USING (auth.ui
CREATE POLICY "own_exploits"   ON exploit_log            FOR ALL USING (auth.ui
CREATE POLICY "own_hss"        ON hidden_skill_signals   FOR ALL USING (auth.ui
CREATE POLICY "own_evolutions" ON profession_evolutions  FOR ALL USING (auth.ui
CREATE POLICY "own_dpq"        ON daily_profile_questions FOR ALL USING (auth.u
CREATE POLICY "read_leaderboard" ON leaderboard_cache    FOR SELECT USING (true
```

---

20. LLM VS APP RESPONSIBILITY SPLIT

Task	Owner
Quest content generation	LLM
Quest type resolution	App
Difficulty calculation	App
Response grading (quality)	LLM
Behavioral signal extraction	LLM (strings only)

Task	Owner
EXP calculation	App
Streak multiplier	App
Level-up check	App
Stat adjustments	App (from LLM signals)
Skill EXP updates	App
Skill evolution naming	LLM
Hidden skill detection	App (pattern matching)
Hidden skill discovery message	LLM
Achievement checking	App
Anti-exploit detection	App
Context compression	App
Behavior summary	LLM (cached every 10 quests)
Memory writes	App
Economy transactions	App
Adaptive state detection	App
Prompt injection	App
Leaderboard calculation	App (Edge Function)
Daily profiling question	LLM (generated) + App (stores + interprets)
Profession evolution check	App
Legendary title (Lv1000+)	LLM
All number changes	App — NEVER LLM

21. TECH STACK

Layer	Tool	Purpose	Cost
Framework	Next.js 14 (App Router)	Web app	Free
Hosting	Vercel	Deployment, edge functions	Free tier

Layer	Tool	Purpose	Cost
Database + Auth	Supabase	Postgres + auth + real-time	Free tier
LLM Primary	Gemini Flash	Quest gen + grading	Free tier
LLM Fallback	Groq (Llama 3.3 70B)	All tasks on Gemini failure	Free
LLM Fallback 2	Hugging Face	Simple tasks only	Free tier
Styling	Tailwind CSS	UI styling	Free
Animation	Framer Motion	Page transitions + animations	Free
State	Zustand	Global state management	Free
Validation	Zod	LLM output schema validation	Free
Code editor	Claude Code (CLI)	AI-assisted development	API usage

22. UI SPECIFICATION

Visual Identity

> Dark cinematic. Synthesis aesthetic. Every screen should feel alive — like a world that knows you're in it.

Color Palette

Element	Color
Background	`#0D0D14`
Surface	`#13131F`
Border	`#2A2A3A`
Amber (CTA, streak, active)	`#C87D2F`
Purple (level, EXP bar)	`#534AB7`
Teal (success, Good grade, hidden skills)	`#1D9E75`
Coral (Creative domain)	`#C85A3A`
Blue (Analytical domain)	`#3A7BC8`
Text Primary	`#E8E6F0`
Text Muted	`#8888A0`

Element	Color
Text Dim	#4A4A60
Gold (Excellent grade)	#FFD700
Red (Fail grade)	#C83A3A
Orange (Bad grade)	#C86A2F

Domain Color Mapping

Domain	Color
TECHNICAL	Purple #534AB7
CREATIVE	Coral #C85A3A
ANALYTICAL	Blue #3A7BC8
SOCIAL	Teal #1D9E75
STRATEGIC	Amber #C87D2F

Typography Rules

```
UI text:              system sans-serif
Skill names:          monospace, amber
Behavior signature:   monospace, amber
Flavor text:          italic, amber-dimmed
Grade reveal mentor:  large, bold, white
No underscores visible anywhere in UI — all skill names display with spaces
Learning style: NEVER shown in UI — backend only
Grade reveal: arrow ↑↓ only — no "+1" number labels
```

Animation Rules (Framer Motion)

```
Page transitions:     350ms ease-in-out
Dashboard → Mark Map: simultaneous slide (left out, right in)
Quest screen:         slide up from bottom (covers entire viewport)
Grade reveal:         cross-fade over quest screen
Event screens:        full page replacements
Card flip:            3D rotateY 180deg, 400ms
EXP bar fill:         spring animation, 600ms
Stat pulse:           scale 1.0 → 1.15 → 1.0, 300ms
Typewriter text:      15ms per character
Skill tags float in:  staggered 200ms, translateY from +20 to 0
Level number:         count-up animation, 800ms
```

```
Particle burst:       8–12 dots, radiate from center, 600ms
Character glow:       CSS filter glow, 2s pulse loop
```

Screen Specifications

SPLASH

```
Background: #0A0A0F full screen
Center: "ZYNTH" in large amber bold + tagline "Synthesize yourself. Reach
your zenith." in muted
3–5 floating amber particles (CSS keyframes, infinite)
1.8s auto-advance → auth check
```

DASHBOARD

```
NAV: 5-tab bottom nav always visible
TOP BAR:
  Left: Level badge (purple pill) + title (amber)
  Right: streak flame + count (amber) or dim dash
  Below: thin EXP progress bar (amber fill, purple track)

DAILY POP-UP (overlay, dismisses after answering):
  Energy check-in: HIGH ⚡ / MID 〰 / LOW 🌙
  Daily profiling question (one question, one tap)
  Both dismissible separately

CHARACTER AREA:
  Center: character silhouette reacting to current state
  Domain color aura based on domain
  Visual states: active/streak/peak/absent/returning

QUEST CARD (amber border):
  Flavor text: italic, amber, 1 line
  Quest title: large white bold
  Difficulty pips: ◆ amber filled, ◇ dim (max 5)
  Estimated time + quest type badge
  Skill tags (spaces not underscores)
  "Begin Quest →" CTA button (full width, amber)
  If generating: shimmer placeholder

MARK ROADMAP (faintly visible behind content, ~15% opacity)
  Becomes dominant on slide to Mark Map tab

MINI STATS (2×2 grid below quest):
  Focus, Consistency, + 2 domain stats
  Name small muted + value large white + ↑↓ arrow if recently changed
```

MARK MAP

```
Slides in from right as Dashboard slides left

ROADMAP (now dominant, full visibility):
  Candy Crush style vertical path
  Each node = one Mark (daily quest day)
  Completed: filled amber circle with checkmark
  Current: pulsing amber outline
  Future: dim #2A2A3A circles
  Every 7th node: larger — Mission node (weekly)
  Path connecting nodes: amber when completed, dim when not
  Currently active node: subtle glow + expand animation

MARK DETAILS (tap a node to expand):
  Quest title, grade received, date, EXP earned
  Tapping current node = starts today's quest

MISSION NODES (every 7th Mark):
  Visually distinct: hexagonal shape, amber + purple
  Shows: mission title, step count, status

SCROLL: vertical, past marks above, future below
```

QUEST SCREEN (full screen, nav hidden)

```
Slides up from bottom covering everything

TOP BAR:
  ← Back arrow (confirm if response typed)
  Timer top right (MM:SS counting up)

QUEST HEADER:
  Type badge + difficulty pips (◆◇◇◇◇)
  Flavor text: italic amber, left vertical accent line
  Quest title: large bold white
  Description: muted
  Skills targeted: small amber chips
  Success criteria: dark box with muted text

INPUT AREA:
  Large multiline textarea
  Background: #13131F, border: #2A2A3A
  Placeholder: "Write your response here..."
  Grows with content, min-height 200px

ACTION ROW:
  [Hint ◎] — tap: slides up bottom drawer with hint
  [Skip ◈1] — tap: confirm dialog, deduct 1 token
  [Submit →] — disabled if empty, amber when active

MULTI-STEP (mission/prestige):
  "Step N of N" indicator with amber dot pips
  Previous steps: collapsed read-only cards above with amber left border
  "Next Step →" until final step
  "Complete Mission →" on final step
```

GRADE REVEAL (full screen, nav hidden)

```
GRADE CARD:
  Appears center, scales in (0.8 → 1.0, spring, 400ms)
  Excellent: amber border + gold text (#FFD700)
  Good:      teal border + teal text
  Basic:     gray border + gray text
  Bad:       orange border + orange text
  Fail:      dark red border, calm tone (no harsh animation)

SEQUENCE (timed):
  0ms:   Card scales in
  400ms: Mentor line fades in (large, bold, white)
  700ms: Feedback paragraph fades in (muted, smaller)
  1000ms: "+N EXP" floats up, animates count from 0
          If streak multiplier > 1.0: show "1.5× streak" below
  1300ms: Stat arrows appear: ↑ teal or ↓ red (no numbers)
  1600ms: Strength chips (teal) stagger in
  1800ms: Improvement chips (amber) stagger in
  2200ms: "Continue →" button appears

On Continue: navigate through queuedEvents then → Dashboard
```

EVENT SCREENS

```
LEVEL UP: Purple theme. Old level fades, new level explodes large.
  New title in amber. Unlocks list. Particle burst.

SKILL EVOLUTION: Teal theme. Old name strikethrough → fades out.
  New name fades + scales in. "Tier N → Tier N+1".

HIDDEN SKILL: Dark full screen. "NEW SKILL DISCOVERED" types slowly.
  2s pause. Skill name appears large in amber. Discovery message types out.
  "+5 XP Cards" slides up. Continue button appears last.

STREAK MILESTONE: Flame emoji scales to 2×. Number counts up.
  Multiplier badge drops in. Token award if applicable.

WEAKNESS OVERCOME: Amber glow pulses on screen edge.
  Old behavior signature shown. New one types over it character by character.
  "+3 permanent to [stat]". "Pattern Broken" badge.
```

PROFILE SCREEN

```
TOP: Level badge + title. Profession large. Domain badge.

CHARACTER CARD:
  Character avatar (if set) or default silhouette
  Character name + gender
  "Customize" button

RADAR CHART (SVG, 6-axis):
  Axes: Focus, Consistency, Adaptability, Execution, Stat5, Stat6
  Scale: 0–25 (allows growth above initial 10)
  Fill: domain color at 25% opacity
  Stroke: domain color full opacity
  Grid rings at 5, 10, 15, 20, 25 (faint #2A2A3A)
  Axis labels with current values

SKILLS: Each skill as a card with name (spaces), tier badge, level bar, level
number.

HIDDEN SKILLS: Discovered (teal) + always 3 ??? slots

STATS GRID: All 6 stats in 2-column grid.

ENGAGEMENT: Streak + 🔥, best streak, quests completed, member since.

ACHIEVEMENTS BUTTON → /achievements page

SETTINGS LINK → /settings
```

LEADERBOARD SCREEN

```
HEADER: "Leaderboard" + Zynth logo

SCOPE SELECTOR (tabs):
  [ Local ] [ State ] [ National ] [ Global ]

CATEGORY SELECTOR (pills):
  [ All Skills ] [ Technical ] [ Creative ] [ Analytical ] [ Social ] [
Strategic ]

TOP 3 (podium style):
  2nd left, 1st center elevated, 3rd right
  Each: avatar placeholder, display name, level, score

LIST (rank 4 onward):
  Rank number | Display name | Domain badge | Level | EXP score

CURRENT USER ROW: Highlighted amber, sticky at bottom of list
  Shows their rank even if not in top visible range

Pull to refresh. Pagination (20 per page).
```

CHRONICLE SCREEN

```
TWO TABS:
  [ My Chronicle ] — personal quest history
  [ Community ]    — recent top responses (anonymized)

MY CHRONICLE:
  Reverse-chronological list of completed quests
  Each entry: title, grade badge, date, EXP, skill tags
  Tap to expand: feedback + mentor line
  Infinite scroll (20 per page)

COMMUNITY:
  Top 5 responses of the week (no names — just responses)
  Domain filter pills
  Purpose: inspiration, not comparison
```

ACHIEVEMENTS SCREEN

```
HEADER: Total achievements unlocked / total visible

RARITY SECTIONS:
  Legendary | Rare | Uncommon | Common

Each achievement: badge icon, name, description (if unlocked), locked
treatment if not
Hidden achievements: shown as ??? with rarity indicator only

MILESTONES SECTION:
  Timeline of major events (first excellent, prestige, hidden skills)
  Based on memory.milestones

HISTORY BUTTON: → Full quest history (Chronicle)
```

---

23. FIRST 10 MINUTES EXPERIENCE

> Retention is won or lost here. Under 5 minutes from first load to hooked.

Time	What Happens
0:00	Splash. "ZYNTH" animates in. Tagline fades. Particles float. 1.8s auto-advance.
0:02	Auth screen. Sign up. Simple.
0:20	Q1 slides in. Feels like a personality quiz. No mention of learning.
1:00	Q1–Q5 complete (~8s per question). "Synthesizing your profile..." loading screen.
1:02	LLM call fires. 2–3s wait. Animated synthesis particles.
1:05	PROFILE REVEAL. Domain badge fades in. Profession appears. Stat bars fill with sound (120ms stagger). Behavior signature types out character by character in amber monospace. Skills float in as tags. Title badge drops last with scale spring.
1:50	FIRST QUEST (calibration). Flavor text is the first thing they read. Difficulty 1. Designed to succeed.
3:30	User submits. "Evaluating..." pulse 1.5s (builds anticipation).
3:32	GRADE REVEAL. Almost always Good or Excellent on calibration. Mentor line hits first in large text. EXP bar animates.
3:45	First level-up NOT triggered here intentionally. EXP bar shows ~25% full — craving completion.
3:50	One stat pulses. ↑ Execution. Small. Real.
4:00	Dashboard. Character glows. Streak flame: 1. Mark Map shows 1 completed node. EXP bar incomplete.
4:10	Daily profiling question pop-up. Feels like a fun bonus question. Not a form.
4:20	Notification prompt (soft): "Want a reminder tomorrow?" One tap.

The result: Under 5 minutes in, the user has a profile that feels like them, one completed quest, a partial EXP bar, a 1-day streak, one mark on their roadmap, and a reason to come back tomorrow.

---

24. ERROR HANDLING & RECOVERY

LLM Failure by Engine

Engine	LLM Fails → App Does
Onboarding	Retry once after 2s. If still fails: show error, answers preserved in local state

Engine	LLM Fails → App Does
Quest Generation	Use fallback quest library (hardcoded quest for user's domain)
Grading	Queue response locally, retry silently on next load
Skill evolution naming	Keep old name, flag for retry
Hidden skill discovery message	Use generic template: "Something that was always in you has finally surfaced."
Behavior summary	Keep cached summary
Daily profile question	Skip silently — no question shown today
Legendary title	Keep previous title, retry on next level-up

Quest Fallback Library (5 quests per domain, hardcoded)

Used when LLM is unavailable. Generic but functional. User always gets a quest.

Domains covered: TECHNICAL, CREATIVE, ANALYTICAL, SOCIAL, STRATEGIC

Each fallback has: quest_title, description, success_criteria, hint, flavor_text. type: 'daily', difficulty: 1, estimated_time: '10 min', steps: []

---

25. FUTURE SYSTEMS

System	Priority	Description
Offline Mode	High	Full offline queue with background sync
Recalibration System	Medium	Re-run Q1–Q5 to detect identity shift
Guild / Team Engine	Medium	Team quests, shared progression, guild leaderboard
Narrative / Lore Engine	Low	Story arcs per profession, worldbuilding, lore unlocks
Seasonal Events	Low	Limited-time quests, community challenges, seasonal marks
Career Path Engine	Low	Real-world outcome mapping, portfolio progression
Analytics Dashboard	Low	Platform-wide engagement tracking (PostHog)
Monetization	Future	Freemium tiers, cosmetics, institutional licensing
Character Visual System	Future	Actual character art that evolves with progression

---

Zynth Blueprint v1.0 — Complete

Synthesize yourself. Reach your zenith.

13 Engines · Next.js + Supabase + Gemini Flash + Vercel Web-first · Claude Code CLI · Dark Cinematic UI

Engines: Onboarding · Quest Generation · Grading · Behavioral Stats · Progression · Hidden Skills · Anti-Exploitation · Context Compression · Memory · Economy · Achievements · Adaptive Intelligence · Continuous Profiling

New Systems: Leaderboard · Character Reactions · Mark Roadmap · Daily Profiling · Profession Evolution via Leaderboard Rank